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Nervous system

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Lecture (1) Synaptic transmission

References:

Guyton and Hall Physiology Textbook of Medical Physiology 13th Edition
Ganong's Review of Medical Physiology, 26th edition

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At the end of this lecture, the student is able to:

1. Describe the main functions of the nervous system and its role in controlling rapid body activities.
2. Differentiate between the components of the Central Nervous System (CNS) and Peripheral Nervous System (PNS) and explain their major functions.
3. Identify the different divisions of the peripheral nervous system including sensory, somatic motor, and autonomic divisions.
4. Define a synapse and explain its functional importance in neural communication.
5. Compare the two types of synaptic transmission: electrical and chemical transmission.
6. Explain the mechanism of chemical synaptic transmission including neurotransmitter release, receptor binding, postsynaptic potentials, and neurotransmitter removal.

NERVOUS SYSTEM:

The nervous system controls the rapid activities.

The endocrine system, regulates the metabolic functions

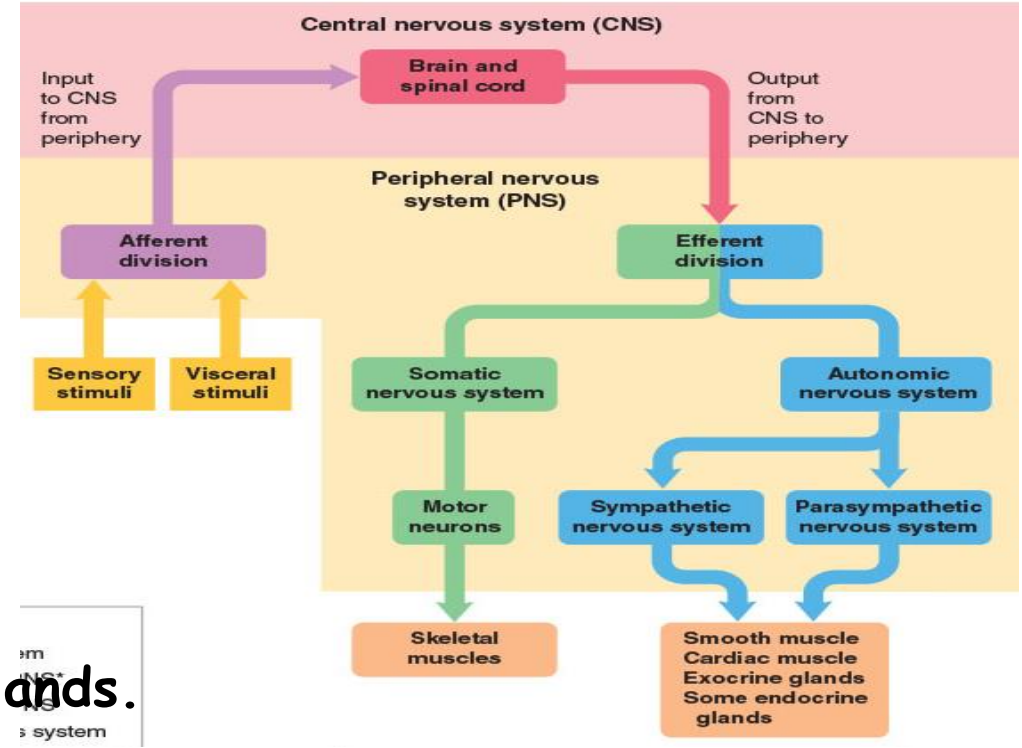
Organization of the nervous system:

A- Central nervous system (CNS):

a. Spinal cord: controls immediate automatic activities (withdrawal reflex; micturition …….)

b. Lower Brain: controls subconscious activities (CVS, RS, posture, emotions)

c. Higher Brain or Cortex: motor & sensory control; thinking, learning, memory & speech.



B- peripheral nervous system

a. Sensory Division: (Afferent Division)

b. Motor Division: (Efferent Division):

- Somatic division → skeletal muscles.
- Autonomic division → smooth & cardiac muscles & glands.

Synaptic transmission:

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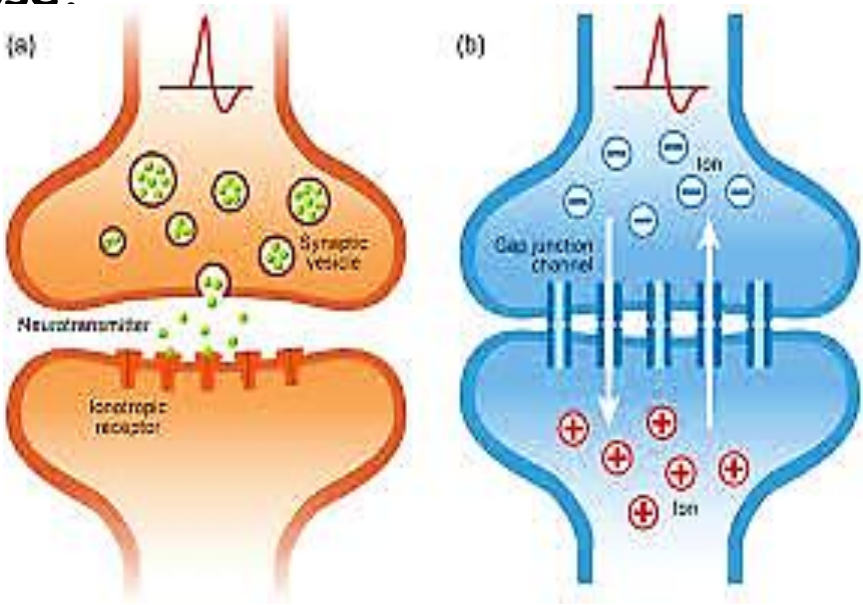
Transmission of impulse from 1 neuron to another along a synapse.



Types of synaptic transmission:

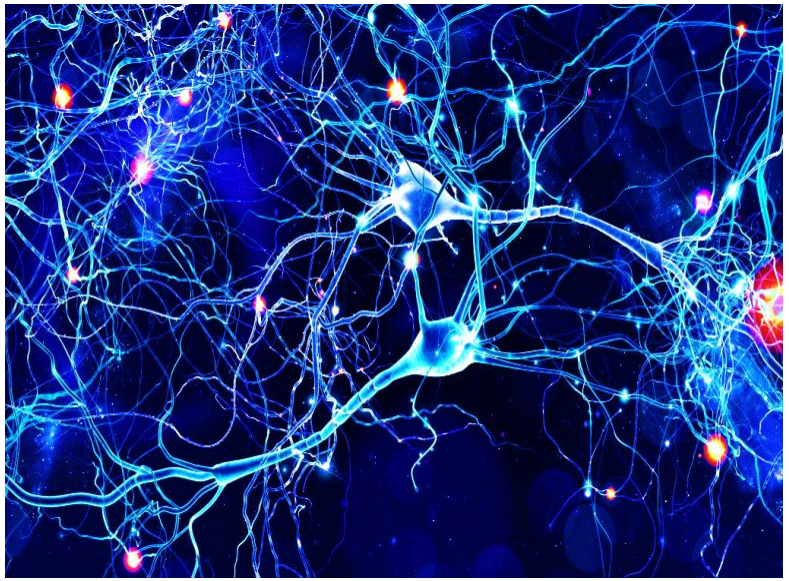
1-Electric transmission →

(rare; through gap-channels; in retina, olfactory bulb)

2-Chemical transmission → (Almost all synapses in human)

What is a synapse?

- junction between an axon terminal of presynaptic neuron and post synaptic neuron.
- Each neuron may form over 2000 synaptic endings
- Each neuron may receive from 200.000 presynaptic terminals.



Functions of synapse:

Synapses permit grading and adjustment of neural activity:

a. Facilitating transmission from one neuron to the next.

b. Blocking transmission from one neuron to the next.

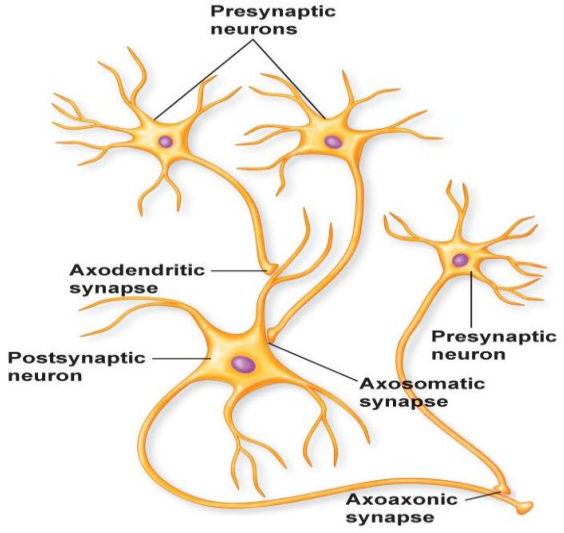
c. Amplification of impulses.

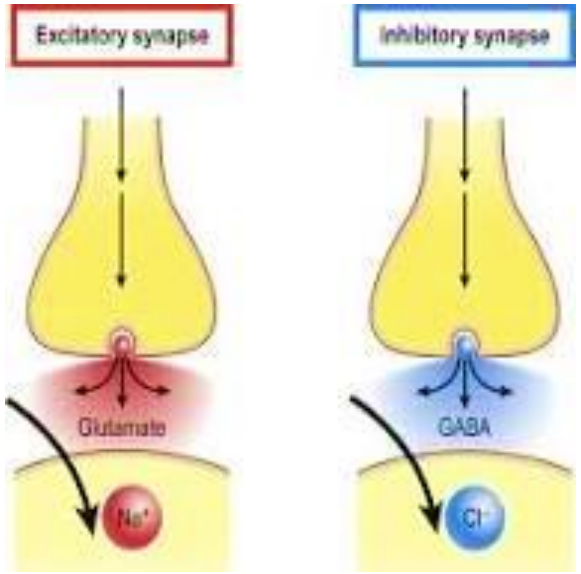
d. Distribution of impulses.

e. Changing one impulse to repetitive impulses.

f. Integration between different types of neurons.

g. Storage of information.





N.B. Synapses are dynamic structures, increasing & decreasing in complexity and number with use and experience.

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Functional Anatomy of a Synapse

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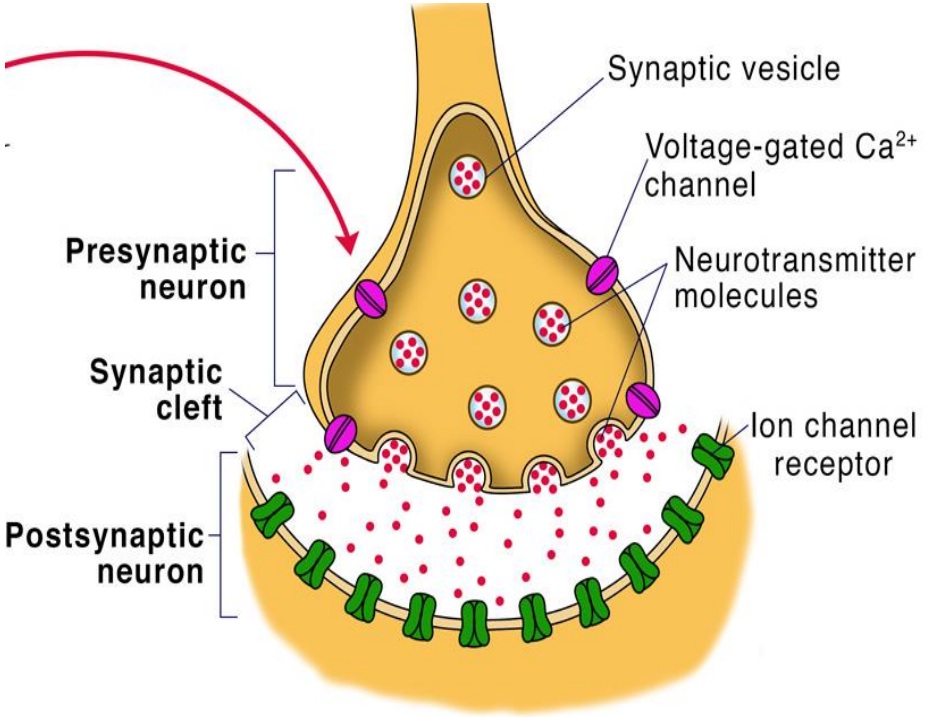
1) Presynaptic Axon Terminal or synaptic knob contains:

a. Mitochondria for ATP synthesis & exocytosis of NT

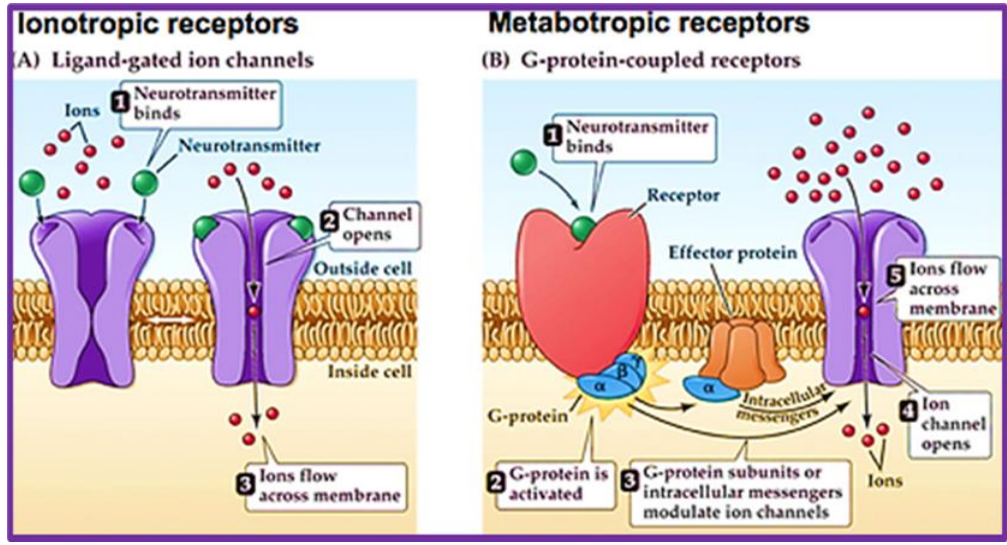
b. Vesicles which contain the synaptic transmitters (NT):

(2) Synaptic cleft: contains ECF rich in Na+, Cl-.

(3) Postsynaptic membrane: contain Receptors:



Ionotropic receptors	Metabotropric receptors
Ligand-gated ion channels	G protein coupled receptors



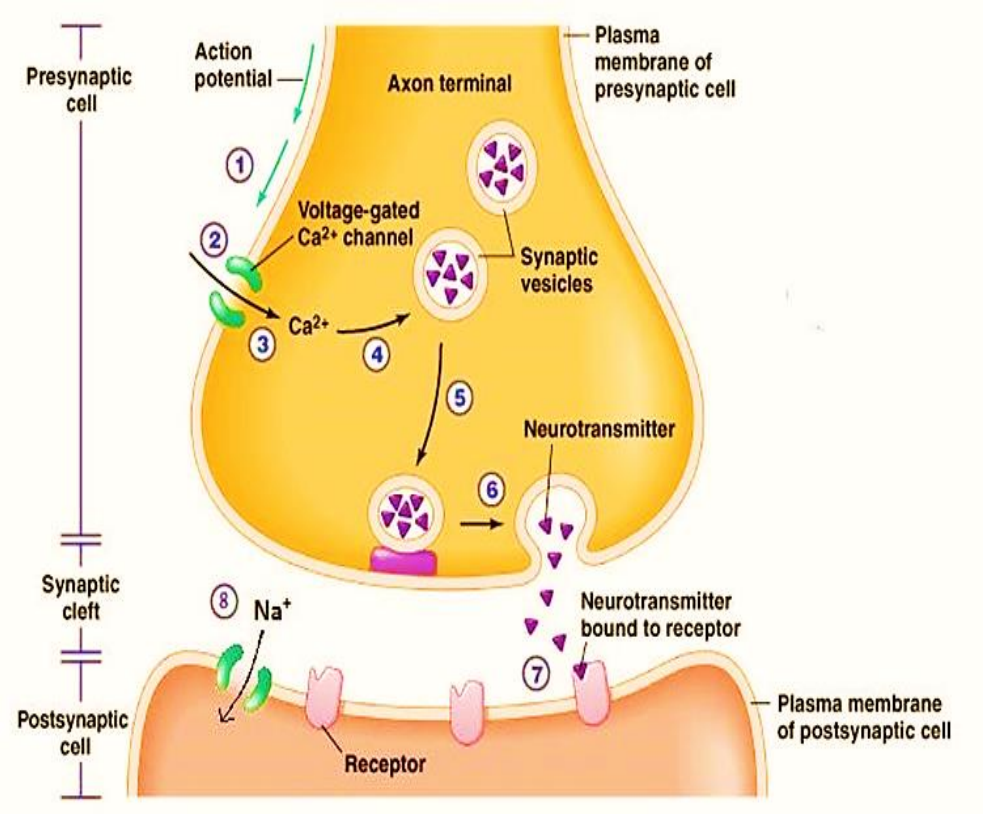
NT-Receptor activate:

- Open or close ion channels
- Activate enzymes
- Activate gene transcription → new proteins.

Mechanism of synaptic transmission

1- Release of chemical transmitter:

- Arrival of action potential in presynaptic nerve → opens voltage-gated Ca++ channels → Ca++ enters the knob with electrochemical gradient.
- Influx of Ca++ → fusion of synaptic vesicles with presynaptic membrane → exocytosis of neurotransmitter.
- Amount of released NT is proportional to Ca++ influx.



2- Binding of chemical Transmitter with its Receptors:

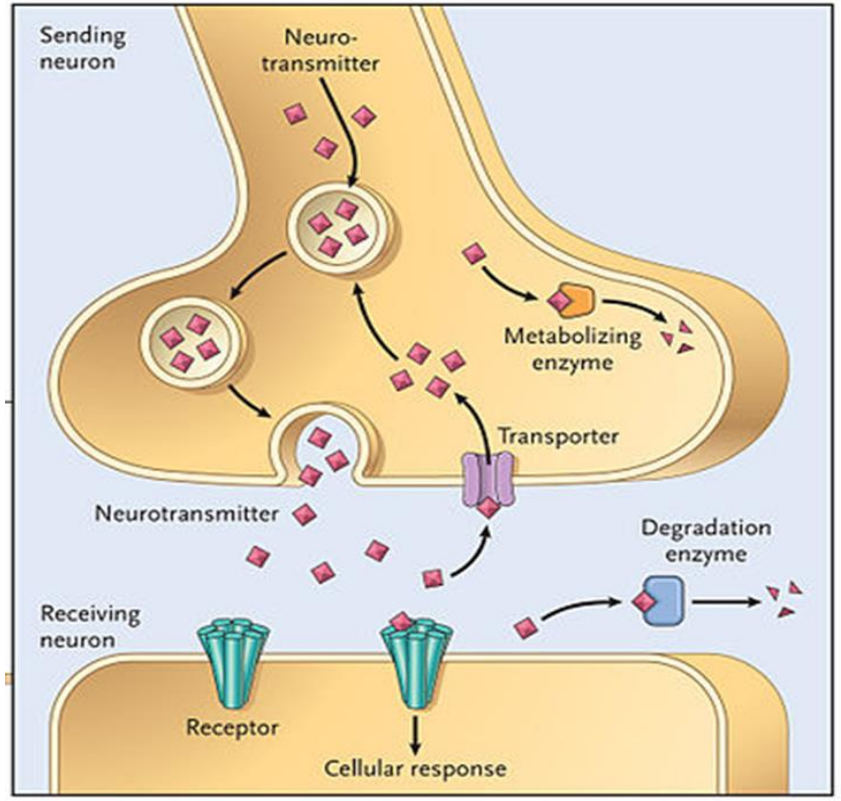
Binding of NT with its receptor → changes the permeability of the postsynaptic membrane to one or more ions.

3- Generation of Postsynaptic Potential:

- Change in ion fluxes → change in membrane potential (according to NT & receptor type):
- less negative → excitatory postsynaptic potential. (EPSP).

Or

- more negative → inhibitory postsynaptic potential. (IPSP).
- Magnitude & duration of EPSP & IPSP depends on the amount of released NT.



4- Removal of Neurotransmitters from the Synaptic Cleft:

- Passive diffusion of NT away from the synaptic cleft (neuropeptide)
- Inactivation of NT by specific enzymes (acetyl choline)
- Active reuptake into the axon terminals (catecholamines, dopamine, serotonin)

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